

Synthetic Glucocorticoid Dexamethasone Inhibits Branching Morphogenesis in the Spiny Mouse Placenta¹

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ABSTRACT

High levels of maternal glucocorticoids during pregnancy can alter the developmental trajectory of some fetal organs. These perturbations are often more profound for the male fetus and have been attributed to passage of glucocorticoids through the placenta. However, the effect of excess glucocorticoids on the placenta itself is less well understood and, particularly, whether this is affected by fetal sex. Expression of genes involved in placental patterning, apoptosis, and nutrient transfer, along with their response to maternal administration of dexamethasone (DEX), has previously been shown to be dependent on fetal sex in the spiny mouse. Here we describe the placental spatiotemporal expression of genes important for branching morphogenesis (*WNT4*, *BMP4*, *GREM1*, *TGFB1*, *KDR*, *VEGFA*). Furthermore, we report that compared to *TGFB1* expression in the female labyrinth, expression of *TGFB1* in the male labyrinth was higher, and earlier peaks in expression levels of *VEGFA* (Day 19 placenta [male] vs. Day 37 labyrinth [female]) and *KDR* (Day 19 placenta [male] vs. Day 20 labyrinth [female]) were observed. Administration of DEX to pregnant dams for 60 h commencing at mid-gestation caused significantly different, sex-related changes in expression of genes that were constitutively different before DEX treatment (e.g., *KDR*, *TGFB1*) and those that were not (i.e., *VEGFA*, *WNT4*). Similarly, some genes which displayed similar expression profiles across gestation for both sexes also showed similar responses to DEX (e.g., *BMP4*), while others did not (i.e., *GREM1*). These results showed that constitutive and glucocorticoid-induced changes in expression of genes involved in branching morphogenesis may be influenced by fetal/placental sex and that fundamental differences exist between a male and female placenta.

developmental origins of health and disease, glucocorticoid receptor, glucocorticoids, placentation, rodents (guinea pigs, mice, rats, voles), stress

INTRODUCTION

It is now accepted that fetal and/or placental exposure to high concentrations of maternal glucocorticoids can lead to

changes in fetal development that may be long lasting and persist into adult life, increasing the occurrence of adult-onset diseases such as hypertension, diabetes, and kidney disease, among others (reviewed in [1, 2–5]); and this risk is somewhat higher for males than for females [6–11]. We and other investigators [12–15] have shown that a relatively short exposure (2.5 days) of pregnant animals to a synthetic glucocorticoid such as dexamethasone (DEX) is sufficient to perturb kidney development, resulting in a decrease in nephron number. This decrease is caused, at least in part, by a reduction in branching morphogenesis during ureteric development so that fewer glomeruli are formed during the fetal phase of kidney development [12]. This outcome was similar in male and female fetuses, but a sex-dependent effect of the glucocorticoid treatment was revealed in adult offspring, where there was significantly poorer survival outcomes for glucocorticoid-exposed adult males than for their female littermates, when exposed to anesthesia and surgery [12].

The role of the placenta in regulating the passage of glucocorticoid from the maternal to fetal circulation has been studied in detail [16–19], but the effect of excess glucocorticoid on the placenta itself is less well understood. We and others have recently shown that exposure to high concentrations of DEX in the spiny mouse at mid-pregnancy resulted in alterations of placental structure and gene expression, with sexually dimorphic changes present at the end of gestation [20–22]. Furthermore, we have shown that in response to DEX administration, placentas from male fetuses were found to have a relative increase in the size of the labyrinth and maternal blood sinusoids within their spongy zones. However, within placentas of female fetuses, both of these parameters were reduced in response to DEX administration [22]. Both of these DEX-induced changes would be expected to affect vascular development of the placenta, and changes observed may be driven by sexually dimorphic DEX-induced expression of branching and vascular mediators. Indeed vasculogenesis mediators within the placenta have been shown to be perturbed following exposure to excess glucocorticoid [20, 23].

Therefore, the aim of the current study was to examine the spatiotemporal expression of genes in the placenta known to have a key role in branching morphogenesis. We tested the hypothesis that DEX treatment at mid-gestation would reduce branching morphogenesis and hence development of the labyrinth region of the placenta, as this region of the spiny mouse placenta undergoes considerable growth and development at this time [24]. We investigated expression of the bone morphogenic protein 4 (*BMP4*), transforming growth factor beta 1 (*TGFB1*), gremlin (*GREM1*), and wingless-type MMTV integration site family member 4 (*WNT4*) genes, all of which have been implicated in branching morphogenesis [25–28] and placentation [29–32], and for which expression is known to be affected by high levels of glucocorticoid [12, 33, 34]. Branching morphogenesis is a tightly regulated process; *TGFB1* and *BMP4* are potent inhibitors of branching

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morphogenesis in a number of different organs including the kidney and lung [12, 35, 36], but their inhibitory actions are counteracted by the presence of promoters of branching such as *GREM1* and *WNT4* [37–42]. Furthermore, we examined mRNA expression of the vascular endothelial growth factor A (*VEGFA*) gene and its receptor, the kinase insert domain receptor (a type III receptor tyrosine kinase) (*KDR*) gene, because of their primary roles in the formation of blood vessels (angiogenesis and vasculogenesis) within the placenta [43], processes also known to be perturbed by high levels of glucocorticoid [20, 23].

The species we used for these studies was the spiny mouse (*Acomys cahirinus*), a rodent species in which offspring are born after a relatively long gestation (39 days), when major organogenesis is complete, so that the formative development of the major organs occurs in utero [44–47]. The developmental trajectory of the placenta [48] and the fact that the fetal and adult adrenal gland synthesizes cortisol and dehydroepiandrosterone in this species [49] provide a small-rodent model which allows both short- (2.5-day) and long-term (2-week) consequences of a brief maternal glucocorticoid pulse to be examined in an in utero environment that is physiologically similar to that of humans.

MATERIALS AND METHODS

Animals

All experiments were approved in advance by Monash University School of Biomedical Sciences Animal Ethics Committee and were conducted in accordance with the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes. The spiny mice (*A. cahirinus*) used in this study were obtained from our own laboratory colony and bred as previously described [44].

Tissue Collection

Dams at Days 15 ($n = 5$), 19 ($n = 4$), 20 ($n = 9$), 25 ($n = 6$), 30 ($n = 7$), 34 ($n = 7$), and 37 ($n = 10$) gestation were killed by cervical dislocation, the uterus was rapidly excised, and placentas and fetuses were removed and weighed. Fetuses >30 days gestational age were sexed by visualization of fetal gonads, and those <30 days were sexed by PCR detection of the presence of lysine (K)-specific demethylase (*Kdm*) gene expression, which has differently sized alleles on the X (5C) and Y (5D) chromosomes [50].

DEX Administration

DEX is a synthetic glucocorticoid which binds preferentially to the glucocorticoid receptor [51] and therefore mimics the effects of endogenous glucocorticoid acting through this receptor. Animal experiments were performed as previously described [12, 22]. Briefly, on Day 20 of gestation, pregnant dams were briefly anesthetized using isoflurane given through a nose cone (4.5% induction, 2.5%–2.8% maintenance dose), and subcutaneously implanted with an osmotic pump (model 1003D; 100 μ l capacity; Alzet), which then delivered DEX (at a concentration of 125 μ g/kg in 75 μ l volume) or saline (75 μ l) over the next 60 h at a rate of 1 μ l/h. On Day 23 of gestation, at the end of DEX or saline administration, dams were allocated to one of two groups, those from which fetal and placental tissues were collected immediately (Day 23: DEX [$n = 6$]; saline [$n = 6$]) or those from which the osmotic pump was removed (again under brief isoflurane anesthesia) and pregnancy was allowed to continue until dams were killed at Day 37 of gestation by cervical dislocation and fetal and placental tissues were collected (Day 37: DEX [$n = 5$], saline [$n = 60$]). Sex determination of fetuses and collection of fetal and placental tissues were performed as previously described [12, 22].

Tissue Preparation

All placentas were divided into two halves across the largest diameter. One-half of the placenta was snap-frozen in liquid nitrogen. For placentas at ≥ 20 days' gestation, one-half was then separated into labyrinth and spongy zone regions by careful dissection prior to freezing. Placental tissue samples frozen in liquid nitrogen underwent total RNA extraction using the RNeasy mini-kit

(Qiagen). Samples (1 μ g of RNA) were then reverse transcribed to form cDNA (reverse transcription system; Promega) and stored at -20°C for later use.

Quantitative Real-Time PCR Analysis

At 15 and 19 days' gestation, it was not possible to visually identify the labyrinth and spongy zones of the placenta, and analysis was therefore done with extracts prepared from the whole placenta. By Gestational Day 20, when it was possible to distinguish and separate the two regions, gene expression analysis was performed separately for each zone. SYBR Green quantitative real-time PCR (qPCR) was used to determine relative levels of mRNA expression of *Bmp4*, *Tgfb1*, *Grem1*, *Wnt4*, *VEGFA*; while a TaqMan multiplex assay was used to analyze *Kdr* expression. Primer sequences (and assay on-demand details) used in this study are shown in Table 1. Optimizing PCR assays were conducted in order to test the specificity of the primer sets, which were found to amplify products of the expected sizes (data not shown). PCR products were then sequenced and confirmed to be the gene of interest using the Basic Local Alignment Search Tool (BLAST; <http://blast.ncbi.nlm.nih.gov>; data not shown). qPCR cycling conditions for all genes consisted of an initial denaturing step of 95°C for 10 min, followed by 40 cycles consisting of 95°C for 15 sec and 60°C for 1 min. A dissociation curve was also performed at the end of each run. Samples were run in triplicate for each gene of interest, and where differences were observed between the sexes or treatment groups, the qPCR was repeated.

Data obtained from qPCR were analyzed as previously described [22]. Briefly, the relative expression (change in cycle threshold [ΔC_T]) for each sample was calculated by subtracting the mean C_T for the housekeeping gene (RNA, 18S ribosomal 1 [*RN18S1*]) [22, 52] from the mean C_T value for the gene of interest. This number was then inserted into the formula $2^{-\Delta C_T}$ to give a final arbitrary expression value. For ontogeny data, the expression value ($2^{-\Delta C_T}$) for each group was divided by the mean $2^{-\Delta C_T}$ value of the Day 37 labyrinth male to give a final value of expression relative to the mean value of the male labyrinth expression at Day 37 for each gene of interest. For the DEX treatment groups, the expression value for each group was divided by the mean $2^{-\Delta C_T}$ of the male saline treatment group of the appropriate age and placental region, to give a final value of expression relative to the mean value of that obtained for the male saline-treated group, for the gene of interest.

Data Analysis and Statistics

All results are represented as means $\pm$ SEM. qPCR data were compiled from placentas of 4–5 fetuses of each sex from different litters for all of the time points shown and from both treatment groups (DEX vs. saline). Ontogeny data generated by qPCR were analyzed using two-way ANOVA for gestational age (P_{TIME}) and fetal sex (P_{SEX}). Where differences at Days 15 and 19 were not apparent between the sexes, these two time points were omitted from repeat two-way ANOVA. Where differences ($P < 0.07$) between groups were revealed in two-way ANOVA, a Bonferroni post hoc test was used to analyze differences between sexes at each time point and also between time points within the same sex. DEX qPCR data were analyzed using two-way ANOVA for treatment (P_{TREAT}) and sex (P_{SEX}).

RESULTS

Gestation-Related Changes of Gene Expression

Early placenta branching genes. Expression levels of *BMP4* and *GREM1* were low in the placenta at Days 15 and 19 (Fig. 1, A and B). Expression of *WNT4* tended to be higher in placentas of female fetuses than in those of male fetuses on Day 15 but not at Day 19 (Fig. 1C). Expression of *TGFB1* did not differ between the sexes in the early placenta (Fig. 1D).

Early placenta vasculogenic genes. Expression levels of *KDR* and *VEGFA* were higher in placentas of male fetuses than in placentas from female fetuses on Day 19 (Fig. 2, A and B).

Definitive placenta branching genes. Expression of *WNT4* in the labyrinth did not change significantly from Day 20 to 37 and was not significantly different for the male and female placentas (Fig. 1C). Expression of *TGFB1* also did not change significantly over this time period but was always significantly higher in placentas from males than in female fetuses ($P_{\text{SEX}} < 0.05$) (Fig. 1D). Expression levels of *GREM1* and *BMP4*, which were greatest in labyrinths at Days 25–30

TABLE 1. Primer sequences for genes of interest.

Gene of interest	Species*	Forward primer sequence (5'–3')	Reverse primer sequence (5'–3')	Reference(s)
<i>RN18S1</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i> , <i>Homo sapiens</i>	ACA CGG ACA GGA TTG ACA GA	CAA CTA GCT CCA CCA ACT AA	[22, 52]
<i>Bmp4</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i>	CGA GCC ATG CTA GTT TGA TAC CT	CCG CGT GGC CCT GAA	[12]
<i>Grem1</i>	<i>Mus musculus</i>	CGT GAC AGA GCG CAA GTA TCT	CTC GTG GAT GGT CTG CTT CA	[12]
<i>Kdm</i>	<i>Mus musculus</i>	TGA AGC TTT TGG CTT TGA G	CCA CTG CCA AAT TCT TTG G	
<i>Kdr</i>	<i>Mus musculus</i>	Mm01222431_m1 [†]	Mm01222431_m1 [†]	[20]
<i>Tgfb1</i>	<i>Mus musculus</i> , <i>Rattus norvegicus</i>	TCG ACA TGG AGC TGG TGA AA	GAG CCT TAG TTT GGA CAG GAT CTG	[12]
<i>VEGFA</i>	<i>Ovis aries</i>	CGA AAG TCT GGA GTG TGT GC	TAT GTG CTG GCT TTG GTG AG	
<i>Wnt4</i>	<i>Rattus norvegicus</i>	CGG CCT TTG TAT ACG CCA TCT	TCC ACT GCT GCA TGC TCT TG	[12]

* Where multiple species are listed, gene sequences of these species were aligned, and consensus regions were used to design primers.

[†] Forward and reverse primers were obtained from Applied Biosystems.

(*GREM1*) and Day 37 (*BMP4*), were not different between placentas of male and female fetuses (Fig. 1, A and B).

Within the spongy zone, expression of *BMP4* was very low and decreased from a peak at Day 20 to lower expression for the remainder of gestation, and this did not differ between a male and female placentas ($P_{\text{TIME}} < 0.05$) (Fig. 1E). Expression of *GREM1* in the spongy zone increased from Day 20 of gestation to reach a peak at term, with no differences between the sexes ($P_{\text{TIME}} < 0.05$) (Fig. 1F). Expression of *WNT4* in the spongy zone was higher in placentas from female fetuses at Day 20 than from those in male fetuses; however, expression in all placentas had decreased to very low levels by Day 25 and remained so for the remainder of gestation (Fig. 1G). Expression of *TGFB1* in the spongy zone decreased from the time of the development of the definitive placenta, but this change occurred earlier in placentas from female fetuses (Day 20) than from those in male fetuses (Day 25) (Fig. 1H).

Definitive placenta vasculogenic genes. Expression of *KDR* in the labyrinths of placentas of female fetuses was higher than in male placentas from Day 20; and at Day 37, the expression of *VEGFA* was higher in the labyrinths of placentas from female fetuses than in those from male fetuses (Fig. 2, A and B).

In the spongy zone, *KDR* expression was higher in the latter half of pregnancy and was not significantly different for males than for females (Fig. 2C). The spongy zone expression of *VEGFA* remained relatively low throughout gestation but decreased (similarly for males and females) after Day 30 of gestation to term (Fig. 2D).

Effects of DEX Treatment on Gene Expression

Both short-term (i.e., Day 23) and longer-term (Day 37) effects of DEX treatment on placental gene expression were observed and are summarized in Table 2. Immediately at the end of DEX treatment on Day 23, expression levels of *BMP4*, *TGFB1*, and *WNT4* were significantly increased in the labyrinth, with similar effects seen for male and female placentas; however, for *TGFB1*, the effect tended to be greater in placentas of female fetuses (Table 2). In contrast, *GREM1* expression was approximately 6-fold increased in the labyrinths of male placentas only by DEX exposure (Table 2). Expression of *KDR* was also increased in placentas of male fetuses following DEX treatment, whereas there was a significant reduction of *KDR* expression in the placentas from female fetuses. In the spongy zone, expression levels of *TGFB1* and *GREM1* were significantly up-regulated by Day 23, immediately after the end of DEX treatment, whereas expression levels of *BMP4* and *WNT4* were significantly

down-regulated in placentas from both male and female fetuses at this time.

Expression of *VEGFA* tended to be increased in the labyrinths of placentas from both male and female fetuses exposed to maternal DEX treatment ($P_{\text{TREAT}} = 0.08$) (Table 2). Expression levels of *VEGFA* and *KDR* in the spongy zone were not affected by maternal DEX treatment at this time (Table 2).

At Day 37 (i.e., 2 weeks after the end of the DEX treatment), expression levels of *BMP4*, *GREM1*, *TGFB1*, *VEGFA*, and *WNT4* were all significantly reduced in the labyrinths of placentas from both male and female fetuses, whereas expression of *KDR* was significantly increased (Table 2). In the spongy zone, expression of *WNT4* was up-regulated in placentas of both male and female fetuses exposed to DEX, while expression of *TGFB1* was significantly down-regulated, and this decrease tended to be greater in the placentas of male fetuses than in those of female fetuses ($P_{\text{INT}} = 0.059$) (Table 2). The DEX treatment resulted in significant down-regulation of *GREM1* expression in the placentas of male but not female fetuses. Expression of *VEGFA* was significantly decreased in both male and female placentas after DEX treatment ($P_{\text{TREAT}} < 0.01$) (Table 2). Expression levels of *BMP4* and *KDR* in the spongy zones of male and female placentas did not differ between DEX and control groups at 2 weeks after the treatment (Table 2).

DISCUSSION

This study has described the spatiotemporal expression of genes in the spiny mouse placenta that, by analogy with other species, are likely to be involved in the development and vascularization of the labyrinth zone of this hemotrichorial placenta. We also examined the spatiotemporal pattern of expression of *WNT4* in placenta for the first time, having made the assumption that *WNT4* was likely to be involved in branching morphogenesis and expansion of the labyrinth, as shown by studies of the developing kidney [12, 25]. We also investigated the immediate and long-term effects of brief exposure to DEX in mid-gestation, as fetal and/or placental exposure to high glucocorticoid concentrations are known to affect fetal and postnatal development (reviewed in [53–55]). We previously described an increase in the expression of genes known to inhibit branching morphogenesis (*BMP4*, *TGFB1*, *GREM1*) in the kidney of the fetal spiny mouse immediately after DEX exposure [12]. Here, we describe similar changes in the expression of these genes in the placenta, and importantly, we report differences for male and female placentas. Specifically, we report that under normal physiological conditions, placentas from male fetuses had higher labyrinthine expression of *TGFB1* and an earlier peak in labyrinthine expression of *VEGFA* and *KDR* than in placentas from female fetuses;

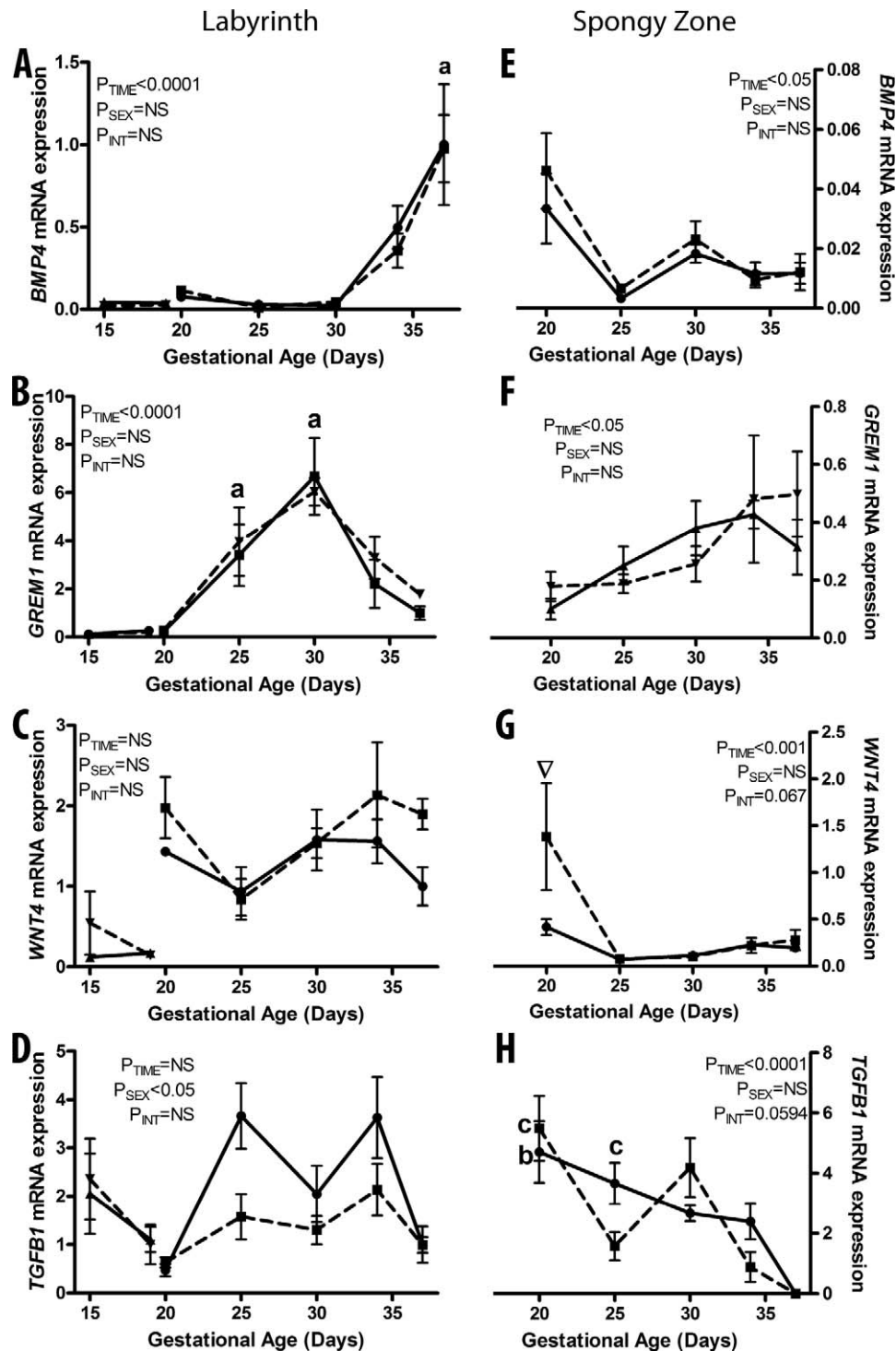


FIG. 1. Changes of branching gene expression with gestation. qPCR assay results show spatiotemporal expression patterns of *BMP4* (A and E), *GREM1* (B and F), *WNT4* (C and G) and *TGFB1* (D and H) in the early placenta (Days 15–19 of gestation [A–D]), along with the labyrinth (left panel [A–D]), and the spongy zone (right panel [E–H]) of the definitive placenta (Days 20–37 of gestation), of male (solid line) and female (dashed line) fetuses. Graph values are means $\pm$ SEM, and all values are ratios of male Day 37 labyrinth mean values for that gene of interest. Two-way ANOVA for treatment (P_{TIME}) and fetal sex (P_{SEX}); P_{INT} refers to an interaction effect between the time (gestational age) and sex effects. Bonferroni post hoc analysis: a, no differences between sexes were observed, but time points that are marked show significant differences from all other gestational ages (A–G); b, time points that are marked show significant differences from all other spongy zone time points for females (H); c, time points that are marked show significant differences from all other spongy zone time points for males (H). ∇ , males and females differed significantly from each other at the particular time points marked.

however, the placentas of female fetuses showed a prolonged period of vascularization. In response to DEX administration, both the placentas from male and female fetuses were found to have an increase in labyrinthine *BMP4*, *WNT4*, *TGFB1*, and

VEGFA expression. In addition to these increases, expression of *GREM1* and *KDR* in the labyrinths of male placentas was increased. Inversely, the expression levels of *KDR* were decreased in placentas of female fetuses.

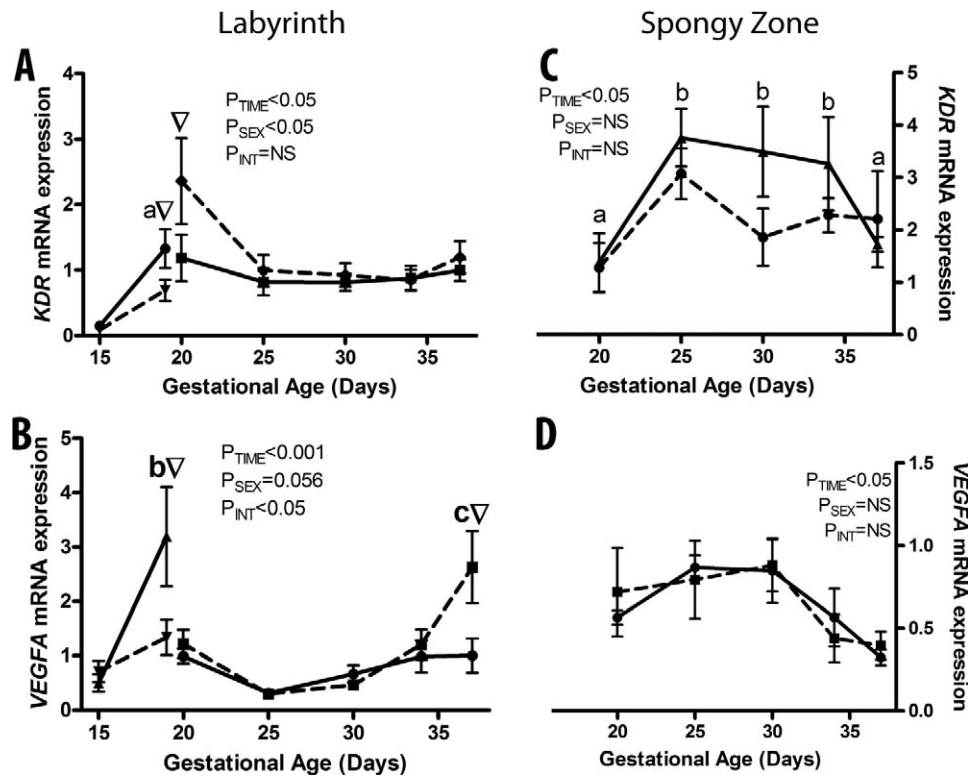


FIG. 2. Changes in vasculogenic gene expression with gestation. With qPCR, the spatiotemporal expression patterns of *KDR* (A) and *VEGFA* (B) in early placenta (Days 15–19 of gestation [A and B]), along with the labyrinth (left panel [A and B]) and spongy zone (right panel [C and D]) of the definitive placenta (Days 20–37 of gestation), of male (solid line) and female (dashed line) fetuses. Graphs values are means $\pm$ SEM, and all values are ratios of male D37 labyrinth mean values for that gene of interest. Two-way ANOVA for treatment (P_{TIME}) and fetal sex (P_{SEX}); P_{INT} refers to an interaction effect between time (gestational age) and sex effects. Bonferroni post hoc analysis: a and b, no differences between sexes were observed, but the time points that are marked are significantly different from all other gestational ages. ∇, males and females differed significantly from each other at the particular time points marked.

Gestation-Related Changes of Gene Expression

The identification of *WNT4* within the placenta has been reported previously [31], but the spatiotemporal pattern of expression of this gene has not been described. The precise role of *WNT4* for placental development is not known, but *WNT2* has been shown by gene knockdown to be involved in the structural development of the labyrinth of the mouse placenta [56]. The *Wnt* family of proteins consists of a number of

locally acting signaling molecules known for their roles in embryonic development and cancer [57–59]. *Wnt* genes affect cell proliferation and differentiation, along with initiating expression in other signaling pathways. Recently, *WNT4* has been shown to be required for normal ovarian development [60], hematopoiesis [61], and neuronal differentiation [62]. Here, most of the *WNT4* expression was found within the labyrinth region of the placenta, with minimal expression in the spongy zone and little expression at all prior to Day 20 of

TABLE 2. Effects of DEX on vasculogenic and branching gene expression.*

Gene of interest	Region	Fold change from saline values at Day 23		Fold change from saline values at Day 37	
		Male (n = 5)	Female (n = 5)	Male (n = 5)	Female (n = 5)
<i>BMP4</i>	Labyrinth	12.1 $\pm$ 3.67 ^a	7.7 $\pm$ 1.2 ^a	0.01 $\pm$ 0.002 ^a	0.01 $\pm$ 0.004 ^a
	Spongy zone	0.14 $\pm$ 0.03 ^a	0.01 $\pm$ 0.01 ^a	0.5 $\pm$ 0.16	0.9 $\pm$ 0.19
<i>TGFB1</i>	Labyrinth	1.6 $\pm$ 0.2 ^a	2.8 $\pm$ 0.6 ^a	0.1 $\pm$ 0.03 ^a	0.34 $\pm$ 0.18 ^a
	Spongy zone	7.7 $\pm$ 1.1 ^a	6.1 $\pm$ 1.5 ^a	0.3 $\pm$ 0.01 ^a	0.56 $\pm$ 0.13 ^a
<i>WNT4</i>	Labyrinth	2.5 $\pm$ 0.3 ^a	2.5 $\pm$ 0.4 ^a	0.4 $\pm$ 0.06 ^a	0.5 $\pm$ 0.2 ^a
	Spongy zone	0.5 $\pm$ 0.09 ^a	0.3 $\pm$ 0.09 ^a	1.1 $\pm$ 0.2 ^a	1.9 $\pm$ 0.2 ^a
<i>GREM1</i>	Labyrinth	6.0 $\pm$ 2.0 ^{a,b}	0.8 $\pm$ 0.1 ^{b,c}	0.2 $\pm$ 0.01 ^a	0.5 $\pm$ 0.3 ^a
	Spongy zone	19.6 $\pm$ 4.2 ^a	6.9 $\pm$ 2.1 ^{a,c}	0.3 $\pm$ 0.1 ^{a,b}	0.9 $\pm$ 0.1 ^b
<i>KDR</i>	Labyrinth	2.1 $\pm$ 0.3 ^b	0.5 $\pm$ 0.1 ^b	2.5 $\pm$ 0.7 ^a	1.5 $\pm$ 0.1 ^{a,c}
	Spongy zone	1.3 $\pm$ 0.2	0.9 $\pm$ 0.3	1.3 $\pm$ 0.36	0.7 $\pm$ 0.1
<i>VEGFA</i>	Labyrinth	1.7 $\pm$ 0.08	1.6 $\pm$ 0.5	0.02 $\pm$ 0.01 ^a	0.06 $\pm$ 0.03 ^a
	Spongy zone	0.9 $\pm$ 0.3	0.5 $\pm$ 0.2	0.4 $\pm$ 0.1 ^a	0.5 $\pm$ 0.1 ^a

* Data are indicated as fold changes $\pm$ SEM from saline values for gestational age, placental region, and fetal sex.

^a Indicates significant fold change ($P_{TREAT} < 0.05$).

^b Indicates significant sexually dimorphic responses ($P_{SEX} < 0.05$).

^c Indicates an interaction effect between gestational age and fetal sex ($P_{INT} < 0.05$).

gestation when the labyrinth/spongy zones cannot be distinguished from each other. Further investigation of the exact cellular location of *WNT4* within the placenta should aid in understanding the function of this particular Wnt gene in placental development.

Sexual dimorphism of expression of many genes was evident from mid-gestation. Within the early male placenta, expression levels of *VEGFA* and its receptor *KDR* were high just before the development of the definitive placenta that occurs at mid-gestation (Day 20), suggesting that vascularization in the placenta precedes the phase of high fetal growth, which commences in this species at ~25 days' gestation [48]. The female placenta may maintain a higher degree of vascularization within the labyrinth throughout gestation, even when fetal growth is at a maximum, as it shows a consistently lower expression of the branching inhibitor *TGFBI* and a consistently higher expression of *KDR*, which can be activated by VEGF and other ligands to promote angiogenesis [63]. This gene expression profile, that is, a higher degree of vascularization, is also suggestive of a greater surface area in the labyrinths of female spiny mouse placentas, an observation which has also been recorded for female human placentas [64]. This fundamental difference between male and female placentas may explain, in part, differences in growth trajectories and responsiveness to changes in maternal nutritional status and health exhibited by male and female fetuses [65, 66].

Within the spongy zone of the placenta, the region where maternal blood is delivered to the labyrinth and where maternal blood is contained in large sinusoids, the expression of the "pro-branching gene" *GREM1* and low expression of the branching inhibitor *BMP4* continued from mid-gestation until near term. The major cell populations in the spongy zone include trophoblasts (both spongiotrophoblast and giant cells), as well as glycogen cells [67], and any or all of these cell populations may increase throughout the latter half of gestation. *GREM1* is expressed in human placental venous endothelial cells during placental development [29]; continued expression of this gene throughout pregnancy in the spiny mouse placenta is consistent with continued development of vasculature, particularly venous, within the spongy zone of the placenta. This could be confirmed by in situ hybridization or immunocytochemical studies that localize *GREM1* within the spongy zone.

Effect of DEX on the Branching Environment within the Placenta

An immediate effect of DEX treatment was increased expression of *BMP4* and *TGFBI*, genes associated with inhibition of branching, a finding consistent with other studies of the placenta and other organs [34, 68–70]. This may result in the formation of fewer, elongated blood vessels similar to those seen within placentas of fetuses born to asthmatic mothers who used glucocorticoid-based medication during pregnancy to manage their respiratory problems [3]. How altered expression of these particular genes might affect blood vessel pattern at this time is not clear, but elsewhere (and discussed further below) we have reported that DEX treatment increases the absolute amount of labyrinth tissue in the placenta at this time [22]. However, by the end of gestation, expression levels of *BMP4*, *TGFBI*, and *VEGFA* were decreased by the DEX treatment compared to those of the saline-treated controls. "The mechanism(s) behind the reversal of the initial increase in gene expression (*BMP4*, *TGFBI*, and *WNT4*) is not known, but we speculate that a negative feedback mechanism may exist

within the placenta, which when faced with aberrant gene expression, promoting vasculogenesis in this case, is initiated. Further work should explore this concept, using gene knockdown/knockin approaches. Results imply that exposure to DEX at mid-pregnancy may ultimately have the effect of increasing branching morphogenesis and vascularization by the end of pregnancy.

As previously mentioned, *GREM1* is involved in the development of placental blood vessels, while *KDR* encodes a receptor for the potent angiogenic factor *VEGFA* [71]. Within the labyrinth, DEX treatment resulted in increased expression of *KDR* and *GREM1* at Day 23 in male fetuses but decreased expression in female fetuses. These sexually dimorphic responses for these two key regulatory genes may explain an observation from our previous study which showed that the DEX-exposed placentas of male fetuses had a larger labyrinth area than those of saline controls at Day 37, while the effect of DEX on the female placenta was the opposite [22]. The functional significance of these changes in gene expression and gross placental morphology are difficult to assess but may indicate that vascular modification takes place early in the male placenta, which then becomes relatively refractory to further change. The downregulation of these genes in the female placenta may provide a mechanism by which the female fetus and placenta is able to reduce exposure to glucocorticoid, thereby reducing further potential negative outcomes, which appear to be more severe for male fetuses [72].

For *VEGFA* and *TGFBI* expressions, DEX exposure has similar effects in both the labyrinth and spongy zones at both Day 23 and 37 of gestation. However, the effect of DEX-exposure on *BMP4* expression in the spongy zone was such that it might be assumed that a "pro-branching" environment was maintained after DEX exposure, unlike in the labyrinth region. This observation suggests that these two placental regions respond quite differently to glucocorticoid and that the labyrinth region is more profoundly affected by the presence of high levels of glucocorticoid during pregnancy.

Taken together, the results of this study indicate that DEX exposure induces changes in the expression of key genes that regulate branching morphogenesis in the placenta and which may affect the early development of the vascular tree within the labyrinth. Although outside the scope of this study, identification of the specific cell types in which the altered gene expression occurs would provide important information regarding the functional outcomes that these alterations may cause, and therefore, further study is warranted. However, other studies have shown that similar changes in gene and/or protein expression are associated with a reduction in the surface area available for exchange of nutrients [23, 73], and similar effects have been seen in the placentas of human fetuses of asthmatic mothers who are treated with inhaled glucocorticoid [3]. The reduction in branching morphogenesis shown here, taken together with our previous findings showing reduced expression of genes important for the partitioning and delivery of nutrients to the fetus (*SLC2A1*, *IGF1* and *IGFIR*), particularly in male fetuses [22], strongly confirms the suggestion of a role for the placenta in the "programming" effect of brief glucocorticoid exposure on fetal growth and development [74].

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